

```
import pandas as pd
import numpy as np
df = pd.read_csv("Q31_bus_punctuality.csv")
print(df.head())
print(df.info())
print("Missing Values:")
print(df.isnull().sum())
df = df.drop_duplicates()
df["scheduled_arrival"] = pd.to_datetime(df["scheduled_arrival"])
print(df["scheduled_arrival"])
df["actual_arrival"] = pd.to_datetime(df["actual_arrival"])
print(df["actual_arrival"])
df["delay_in_minutes"] = (df["actual_arrival"] - df["scheduled_arrival"]).dt.total_seconds() / 60
print(df["delay_in_minutes"])
df["punctuality"] = np.where(df["delay_in_minutes"] <= 5, "On Time", "Delayed")
print(df["punctuality"])
route_delay = df.groupby("route_number")["delay_in_minutes"].mean()
print("Average Delay by Route:")
print(route_delay)
print("Routes with high average delays:")
print(route_delay[route_delay > 10])
print("Final Dataset:")
print(df)
```

	trip_id	route_number	scheduled_arrival	actual_arrival	passenger_load	\
0	BUS0001	R3	2026-01-18 00:42	2026-01-18 00:37	24	
1	BUS0002	R5	2026-05-22 17:09	2026-05-22 17:30	51	
2	BUS0003	R3	2026-02-18 08:54	2026-02-18 09:13	74	
3	BUS0004	R2	2026-02-07 05:21	2026-02-07 05:17	69	
4	BUS0005	R4	2026-04-11 15:34	2026-04-11 15:44	32	

	traffic_level	weather
0	Low	Clear
1	High	Rain
2	Moderate	Rain
3	Low	Clear
4	Moderate	Clear

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 123 entries, 0 to 122
Data columns (total 7 columns):
#   Column          Non-Null Count  Dtype
---  -
0   trip_id         123 non-null   object
1   route_number    122 non-null   object
2   scheduled_arrival 123 non-null   object
3   actual_arrival   123 non-null   object
4   passenger_load   123 non-null   int64
5   traffic_level    123 non-null   object
6   weather         121 non-null   object
dtypes: int64(1), object(6)
memory usage: 6.9+ KB
None
Missing Values:
trip_id      0
route_number 1
scheduled_arrival 0
actual_arrival 0
passenger_load 0
traffic_level 0
weather      2
dtype: int64
0    2026-01-18 00:42:00
1    2026-05-22 17:09:00
2    2026-02-18 08:54:00
3    2026-02-07 05:21:00
4    2026-04-11 15:34:00
...
115   2026-02-12 05:26:00
116   2026-05-10 05:50:00
117   2026-05-28 04:35:00
118   2026-03-15 08:19:00
119   2026-06-16 22:20:00
Name: scheduled_arrival, Length: 120, dtype: datetime64[ns]
0    2026-01-18 00:37:00
1    2026-05-22 17:30:00
2    2026-02-18 09:13:00
3    2026-02-07 05:17:00
4    2026-04-11 15:44:00
...
```

```
115 2026-02-12 05:40:00
116 2026-05-10 06:15:00
117 2026-05-28 04:59:00
```